

Chapter 5 Answer Key

Introduction to Statistics — MATH& 146

Section 5.2 Exercises

- 5.2.1. The confidence interval is $48\% \pm 3\% = (45\%, 51\%)$.

In plain language: the polling firm used a procedure that, if repeated across many different random samples, would produce an interval containing the true approval rating 95% of the time. This particular interval, (45%, 51%), was produced by that procedure. We do not know whether this specific interval contains the true rating, but 95% of intervals built this way do.

- 5.2.2. The intervals would be **narrower**. A 90% CI uses $z^* = 1.645$ instead of $z^* = 1.96$; the smaller critical value produces a shorter interval. As a consequence, **more of the intervals would miss** μ — specifically, about 10% would miss instead of 5%. Narrower net, more misses: there is always a tradeoff between width and reliability.

- 5.2.3. **False**. Once the interval (42, 58) has been computed, μ is a fixed (unknown) number; it either lies inside (42, 58) or it does not. There is no probability to assign. The 95% refers to the *procedure*: if we repeated the sampling and interval-construction process many times, 95% of the resulting intervals would contain μ . It is not a statement about any particular value of μ such as $\mu = 50$.

Section 5.3 Exercises

- 5.3.1. $\bar{x} = 1,182$ hr, $\sigma = 80$ hr, $n = 50$.

Standard error: $SE = 80/\sqrt{50} \approx 11.31$ hr.

- (a) **90% CI**: $z^* = 1.645$.

$$ME = 1.645 \times 11.31 \approx 18.6 \text{ hr.} \quad 1,182 \pm 18.6 \Rightarrow (1,163, 1,201) \text{ hr.}$$

- (b) **99% CI**: $z^* = 2.576$.

$$ME = 2.576 \times 11.31 \approx 29.1 \text{ hr.} \quad 1,182 \pm 29.1 \Rightarrow (1,153, 1,211) \text{ hr.}$$

- (c) The 99% interval is wider (58.2 hr wide vs. 37.2 hr wide for the 90% interval). The 90% interval gives more *precision* — a narrower range of plausible values — but at the cost of lower confidence that the interval actually contains μ .

- 5.3.2. Required sample size for a 95% CI with $ME \leq 15$ hr, $\sigma = 80$ hr:

$$n \geq \left(\frac{z^* \cdot \sigma}{E} \right)^2 = \left(\frac{1.96 \times 80}{15} \right)^2 = (10.453)^2 \approx 109.3.$$

Round up: $n \geq 110$ bulbs.

5.3.3. A 95% CI must capture the central 95% of the standard normal distribution, leaving 2.5% in each tail; searching the z-table for a left-tail area of 0.975 gives $z^* = 1.96$. A 90% CI captures only the central 90%, leaving 5% in each tail, so we search for a left-tail area of 0.95 and find $z^* = 1.645$. Higher confidence requires a larger critical value and therefore a wider interval.

Section 5.4 Exercises

5.4.1. $n = 500$, $x = 285$ voters favoring Candidate A.

- (a) $\hat{p} = 285/500 = \mathbf{0.57}$.
- (b) Success-failure condition: $n\hat{p} = 500(0.57) = 285 \geq 10 \checkmark$ and $n(1 - \hat{p}) = 500(0.43) = 215 \geq 10 \checkmark$. The normal approximation is valid.
- (c) **95% CI:**

$$SE = \sqrt{\frac{0.57 \times 0.43}{500}} = \sqrt{0.000490} \approx 0.0221. \quad ME = 1.96 \times 0.0221 \approx 0.0434.$$

$$0.57 \pm 0.0434 \Rightarrow (\mathbf{0.527}, \mathbf{0.613}).$$

- (d) **Yes**, we can conclude the majority will vote for Candidate A. The entire interval (0.527, 0.613) lies above 0.50, so at the 95% confidence level the true proportion is greater than 50%. Even the lower bound of the interval, 52.7%, represents a majority.

5.4.2. The midpoint of the interval is the point estimate:

$$\hat{p} = \frac{0.34 + 0.46}{2} = \mathbf{0.40}.$$

The margin of error is half the width of the interval:

$$ME = \frac{0.46 - 0.34}{2} = \mathbf{0.06}.$$

5.4.3. At fixed n , $SE(\hat{p}) = \sqrt{\hat{p}(1 - \hat{p})/n}$. The function $\hat{p}(1 - \hat{p})$ is maximized when $\hat{p} = 0.5$, where $\hat{p}(1 - \hat{p}) = 0.25$. At $\hat{p} = 0.1$, we get $\hat{p}(1 - \hat{p}) = 0.09$, which is considerably smaller.

For example with $n = 100$: $SE = 0.050$ at $\hat{p} = 0.5$ versus $SE = 0.030$ at $\hat{p} = 0.1$. Estimation is most uncertain when $\hat{p}$ is near 0.5, because that is when individual outcomes are most evenly split and therefore most unpredictable.

Chapter 5 Review

1. $\bar{x} = \$28,400$, $\sigma = \$6,200$, $n = 64$.

Standard error: $SE = 6,200/\sqrt{64} = 6,200/8 = \775 .

- (a) **90% CI:** $ME = 1.645 \times 775 = \$1,274.88 \approx \$1,275$.

$$28,400 \pm 1,275 \Rightarrow (\mathbf{\$27,125}, \mathbf{\$29,675}).$$

- (b) **95% CI:** $ME = 1.960 \times 775 = \$1,519$.

$$28,400 \pm 1,519 \Rightarrow (\mathbf{\$26,881}, \mathbf{\$29,919}).$$

(c) **99% CI:** $ME = 2.576 \times 775 = \$1,996.40 \approx \$1,996$.

$$28,400 \pm 1,996 \Rightarrow (\$26,404, \$30,396).$$

(d) As confidence increases from 90% to 95% to 99%, the interval grows wider: from \$2,550 wide to \$3,038 wide to \$3,992 wide. The tradeoff is precision versus certainty: higher confidence that μ is inside the interval comes at the cost of a less informative (broader) range of plausible values.

2. Required sample size for a 95% CI with $ME \leq 1$ hour, $\sigma = 5$ hours:

$$n \geq \left(\frac{1.96 \times 5}{1} \right)^2 = (9.80)^2 = 96.04.$$

Round up: the university must sample $n \geq 97$ students.

3. $n = 400$, $x = 148$ adults reporting ≥ 8 hours of sleep.

(a) $\hat{p} = 148/400 = \mathbf{0.37}$.

Conditions: $n\hat{p} = 400(0.37) = 148 \geq 10 \checkmark$ and $n(1 - \hat{p}) = 400(0.63) = 252 \geq 10 \checkmark$.

(b) **95% CI:**

$$SE = \sqrt{\frac{0.37 \times 0.63}{400}} = \sqrt{0.000583} \approx 0.0241. \quad ME = 1.96 \times 0.0241 \approx 0.0473.$$

$$0.37 \pm 0.0473 \Rightarrow (\mathbf{0.323}, \mathbf{0.417}).$$

(c) **Interpretation:** We are 95% confident that the true proportion of adults who get 8 or more hours of sleep per night is between 32.3% and 41.7%.

(d) The agency's claim of 40% is **consistent** with the interval: $p = 0.40$ lies inside $(0.323, 0.417)$, so 40% is one of the plausible values for the true proportion at the 95% confidence level. The data do not contradict the claim.

4. A 95% CI for a mean is $(44.2, 55.8)$.

(a) $\bar{x} = (44.2 + 55.8)/2 = \mathbf{50.0}$. Margin of error $= (55.8 - 44.2)/2 = \mathbf{5.8}$.

(b) A 90% CI from the same data uses $z^* = 1.645$ instead of $z^* = 1.96$. Since ME scales proportionally with z^* :

$$ME_{90} \approx 5.8 \times \frac{1.645}{1.96} \approx 4.87. \quad \text{Approximate 90\% CI: } (\mathbf{45.1}, \mathbf{54.9}).$$

The 90% interval is narrower, as expected.

(c) The correct statement is: "I used a procedure that, in the long run, produces intervals containing the true mean 95% of the time. This interval, $(44.2, 55.8)$, was produced by that procedure." The true mean μ is a fixed number; once the interval is computed, μ either is or is not inside it. There is no probability left to assign to this specific interval.

5. Model response (writing prompt):

The Central Limit Theorem guarantees that $\bar{x} \sim N(\mu, \sigma/\sqrt{n})$ for large n , regardless of the shape of the original population; without this result, we would have no reason to believe that the standard normal table describes the behavior of sample means, and we could not translate a desired confidence level into a critical value z^* . Because the CLT gives us the sampling distribution of $\bar{x}$, we can find the exact range $[\bar{x} - z^* \cdot \sigma/\sqrt{n}, \bar{x} + z^* \cdot \sigma/\sqrt{n}]$ that covers μ in exactly the desired percentage of samples. If the CLT did not hold — that is, if $\bar{x}$ were not approximately normal — the z-table would give the wrong probabilities, the stated confidence level would be meaningless, and no valid confidence interval could be constructed without knowing the exact population distribution.